Cloze, but not the same: Response entropy depends on language and language dominance

Introduction: Cloze tests are commonly used in psycholinguistics to estimate sentence constraint. In these tasks, participants complete sentence contexts with the first word(s) that come to mind, and the distribution of responses is then analyzed to quantify how constraining a sentence context is. Such measures are often used to define high versus low constraining conditions in experimental studies on prediction, for instance. For cloze norms to be meaningful, however, they must reflect the linguistic population under investigation. The utility of cloze data therefore depends on how closely the pre-test sample matches the participants in the main experiment. Yet, there is limited evidence on how differing sociolinguistic profiles influence such measures. This study directly tests whether sentence constraint varies as a function of language (Experiment 1) and language dominance (Experiment 2).

Methods: We conducted a cloze test on 330 sentence contexts across different language groups. Participants wrote the first continuation that came to mind for each sentence, and lexical sentence constraint was quantified as within-group response entropy. In Experiment 1, 60 French-dominant and 60 Spanish-dominant speakers completed translated versions of the same sentence materials in French and Spanish, respectively. In Experiment 2, the same contexts were presented in French to three groups: 60 French-dominant (same as in Experiment 1), 60 Spanish-dominant, and 60 English-dominant speakers. We then assessed how closely group pairs aligned in sentence constraint computing Spearman correlations. To further interpret the observed correlations, we obtained a distribution of observed correlations through bootstrapping and developed two additional correlation benchmarks tailored to our data: a distribution of correlations representing null alignment, obtained through permutation tests, and a distribution representing maximum alignment, derived from split-half iterations.

Results: In Experiment 1, the correlation of response entropy scores between Spanish- and French- dominant speakers was .55, but the bootstrapped distribution was distant from the maximum alignment distribution (d = -3.77) (Figure 1). In Experiment 2, the correlation between French- and Spanish- dominant speakers was .53, and the correlation between French- and English- dominant speakers was .60. In both cases, the bootstrapped correlation distribution was distant from the maximum alignment distribution (d = -5.19 and d = -3.18, respectively) (Figures 2 and 3). Nevertheless, Spanish- and English-dominant speakers achieved a correlation of .71, their bootstrapped correlation distribution overlapping the distribution of maximum alignment (d = -0.20) (Figure 4).

Conclusion: Since presenting the same translated sentence contexts to dominant speakers of different languages resulted in only moderate alignment, our findings suggest that sentence expectations are shaped by language-specific usage and cultural factors associated with high proficiency and dominance. Moreover, differences in constraint scores between dominant and non-dominant speakers highlight the role of language dominance in shaping predictive processing. In contrast, participants' most dominant language did not affect sentence constraint in a shared non-dominant language, suggesting minimal L1 influence on L2 lexical expectations (at least with close L1s like Spanish and English). This study provides the first empirical examination of how sociolinguistic variables impact sentence constraint, contributing to theoretical proposals of L1 and L2 prediction, and supporting more informed practices for designing language prediction studies across diverse speaker populations.

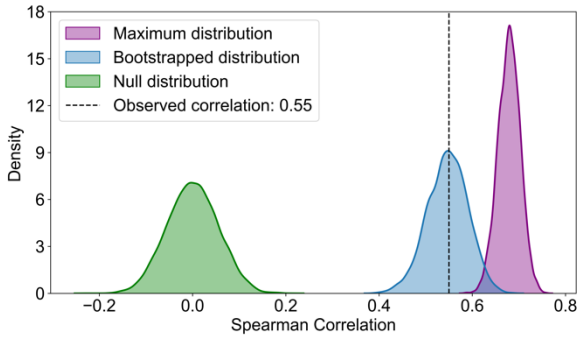

Figure 1. Alignment between French- and Spanish-dominant speakers completing French and Spanish sentences, respectively.

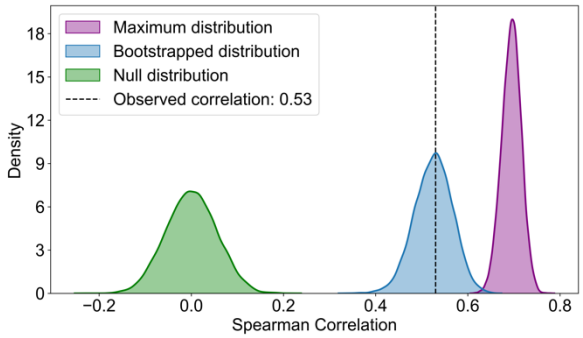

Figure 2. Alignment between French- and Spanish-dominant speakers completing French sentences.

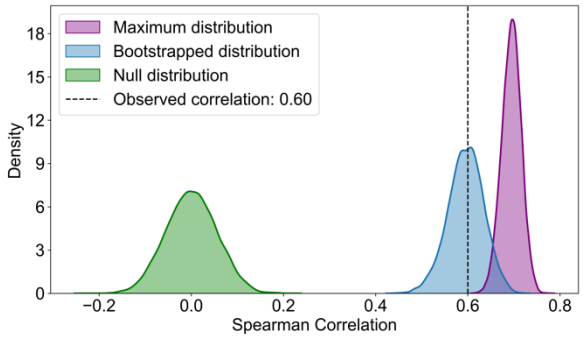

Figure 3. Alignment between French- and English-dominant speakers completing French sentences.

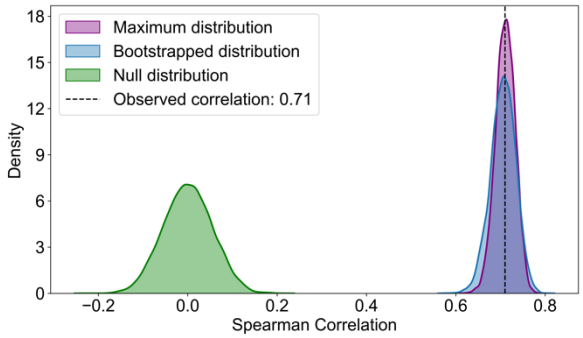

Figure 4. Alignment between Spanish- and English-dominant speakers completing French sentences.